Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · CLASS 30 Beams Numerical

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 30 Beams Numerical

1 CLASS 30 Beams Numerical

Study focus: CLASS 30 Beams_Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000086 · EV0000087

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is question 5/101.

Key details

- Question is determine the reactions at A and B for the beam subjected to the uniform load distribution.
- Look at this question.
- There is a beam AB.
- The name of the beam is simple beam because one end of the beam is having pin connection, other end is having roller, and we need to find the support reactions at A and B.
- Since it is a pin connection, there will be two reactions, one along x direction, one more along y direction.
- We need to find what is ax and what is a y.
- Next, at B, since it's a roller, there will be only one reaction that is reaction at B R B that we have to calculate.
- These three reactions will be because of this loading on the beam.
- What is the name of this loading?
- Remember it's UDL uniformly distributed loading that we call it as rectangular loading also.
- First thing is always we have to convert this UDL into a point load because in our free body diagram we want all the forces to be point load or concentrated loads.
- Then how to convert this UDL into a point load?
- What is the formula?
- Do you remember?
- The formula is the resultant of this UDL R will be equal to W into L.
- What is W intensity of this loading.

- And L is span over which it is acting.
- Here what is W?
- W is intensity that is 6 kilo Newton per meter and what is L?
- Span that is 300 meter millimeter.
- Just take care of the conversion.
- Whenever we want to draw the free body diagram, the first step is we need to project all these important points vertically downward just like this.
- This is the point where the load is ending.
- Just draw a vertical line, then draw a vertical line passing through B.
- Either start from left end of the beam or from the right end of the beam so that you will not miss any of the forces in the free body diagram.
- I am going to start from this end of the beam and I'm going towards right side.
- Look at the beam here.
- At EA it is having a pinned connection.
- Pinned connection means two reactions.
- This is the assumption we are going to make.
- Remember, if we get the answer as positive, our assumption is correct.
- If you are getting the answer as negative, then we have to change the direction of the arms.
- That's it.
- No need to worry.
- Next we have UDL here.
- The intensity of the point load is 1.8 kilonewton, so 1.8 kilonewton here.
- But where exactly this will be acting?
- Remember the formula for U UDL it will be at a distance of L by 2 from here and L by 2 from here.
- Since L is 3000 it will be equal to 150 mm and 150 mm.
- Next we don't have any loading from in between this three hundred millimeters, so no need to do anything here.
- Then come to B at B since it's the roller support.
- The roller will offer only one reaction that is.
- Once we are done with the free body diagram of a beam, we have to apply the conditions of equilibrium.
- Look at the forces here.
- They constitute general force system.
- There is only one horizontal reaction that is Ax.
- We don't have any other horizontal force or reaction.

Example or application from the lecture

- Let's get into the solution part.
- Since the load is given per meter and this is in millimeter, convert from meter to millimeter or millimeter to meter.
- Once we know the resultant of the loading let's start with the free body diagram of the bean A B.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

Core explanation · 05:00–09:59

So we are done with the first one.

Key details

- One is a y.
- This one and one more is RB this one.
- To avoid that I'll go I'll start with the third condition of equilibrium that is moment about any point zero.
- Remember this is sum of the moment of all the forces about A should be 0.
- Next, move in this direction towards right.
- After this, the next force is 1.8 kilonewton.
- Apply the right hand thumb rule.
- Once you apply the right hand thumb rule here, your finger thumb is going inward.
- If the thumb is going inward, that indicates it is.
- Clockwise and in one more way if you are moving in this direction to reach here, you have to take this turn.
- This is the shortest rule.
- Clockwise means negative.
- That's what is written here.
- We are done with the moment of 1.8 kilonewtons.
- From here to here we don't have any forces.
- Then come to RB.
- RB is acting in the counterclockwise direction.
- forces R B into perpendicular distance.
- What is the perpendicular distance from point B to A?
- It is 150 plus 150 plus 300.
- Once you simplify this, you are going to get the value of RBS positive plus 0.45 kN.
- What the plus sign indicates our assumption is correct.
- We assume the force RB is upward.
- That is what this plus sign indicates.
- We are done with the reactionate roller B.
- Next we are left with only one more unknown that is AY.
- Then how to find AY?
- We have one more condition of equilibrium that is sigma f of y is equal to 0.
- Apply this sigma f y is equals to 0.
- AY is upward.
- Then 1.8 is downward.
- RB is upward.
- That should be equal to 0.
- We know the value of RB that is 0.
- 45 do the substitution here.

- Then you simplify, you'll get the value of a1.
- You'll get the value of Ay as 1.35 kN and the answer is positive.
- That indicates our assumption is correct.
- Remember, since it is a pin connection, what we can do here is we can write the reaction at A as under root of ax square plus a y square.
- Ax is 0, so reaction at A will be equals to directly AY.
- That's it.
- This is the next question.
- 5 slash 103.
- Calculate the reactions at A and B for the beam loaded as shown.
- Do you remember the name of this beam?
- It is called as combination or overhanging beam.
- Combination is because it is having pin connection here and it's a roller.
- Overhanging is this part of the beam, this one meter is extended beyond the support B roller support B.
- Hence it is called as overhanging beam.
- What is the name of this loading?
- It is the combination of triangular and rectangular loading.
- Then how to divide this loading?
- We know if we draw a line ear, horizontal line passing through this, it will be a rectangle down and a triangle ear.

Example or application from the lecture

- Next, if you consider second condition sigma f y 0, we have two unknowns along y direction.
- If you consider Ax and AY, these two are acting at the point A itself.
- Consider it as plus.

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Core explanation · 10:00–14:59

Let's see how we can draw the free body diagram for this.

Key details

- First thing is we have to divide this into rectangle and triangle.
- To divide that into rectangle and triangle, just draw a horizontal line passing through this.
- For the time being, forget about this triangle.
- Is it nothing but rectangular loading are UDL uniformly distributed loading?
- Then how to find the resultant of UDL?
- For that we know the formula W into L.
- W stands for intensity of the loading and L stands for the span over which the load is acting.
- Here W is right now 400 because that is the height of this rectangle.
- And the span L is nothing but the distance over which it is acting.
- 6 plus 1 plus 1 that is 8 meters.

- Since this is also in meter and the distance is also in meter, no need to worry about the conversion part.
- So total the resentment of this first loading is 3.2 kilonewtons.
- The second one is triangle.
- Just forget about the rectangle.
- We know the formula R is equals to half into base into height.
- Base is here A 6 meters because the triangle is having a base steel here, that's it.
- What is the height of this triangle?
- Total is eight hundred.
- Then how to find this height of a triangle?
- Total eight hundred minus height of the rectangle.
- You will get this height.
- This is W L by 2.
- And one more thing.
- When you talk about R1, that is the resultant of this rectangular loading, where exactly will that be active?
- L by 2 distance with respect to a and L by 2 with respect to this end.
- Nothing but 8 by 2 meters.
- Then where will be the resultant of this triangle with respect to this side we know L by 3.
- If you are talking from this side, it will be 2L by 3.
- That's why I have written it as next.
- L by 3 with respect to A.
- 6 by 3 is equal to 2 meters.
- Next, we can now we can start with the free body diagram of the beam.
- Need to draw the free body diagram.
- Remember we need to project all these points vertically downward.
- This will be the beam AB.
- Start from any one of the end, either from this end, or from this end.
- What is the name of this connection?
- It is a pin connection.
- This is up to you the direction.
- We are done with this.
- Next what do we have?
- We have R2 from here 1.2 kilonewton at a distance of 2 meters.
- We are done with the resultant of the second one.
- Where is the resultant of the first one with respect to 4 meters.
- 4 meters means after R2 we are going to get that.
- After this, we are done with the resultant of the loadings.
- Next move in this direction.
- We don't have any other loading, but here at a distance of 7 meters with respect to A, we have a roller support.
- We are done with the free body diagram of the beam.
- Once we are done with the free body diagram of the beam, next thing is we need to apply the conditions of equilibrium.

- What is the first condition of equilibrium?
- Sigma F is equal to 0.
- Since there is no horizontal force acting on the beam, Ax will be straight away.
- it will be equal to 0.
- I'll start with the moment equation because if you apply sigma f y 0, we have two unknowns a y then r b.
- Just to avoid that I am going to apply moment about E as 0.

Example or application from the lecture

- Just consider this rectangle.
- For when you consider a triangle, you no need to consider the rectangle.
- Consider that as it is one meter.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 15:00–19:59

Once you apply moment about E as 0, look at this at A we have two forces Ax and AY.

Key details

- Since these two are acting at the same point, moment of these two forces will be zero.
- That's the reason why I'm not considering Ax and AY.
- More in this direction, which is the next force we have?
- 1.2 kilonewton downward.
- Don't look whether it is downward or upward.
- When we are talking about moment you should think whether it is.
- Making clockwise rotation or anti-clockwise rotation with respect to point under consideration.
- If your thumb is going inward, so it is clockwise.
- Since it is clockwise it should be written as minus into what is force here?
- What is the perpendicular distance?
- Next comes the second force that is 3.2 kilonewton.
- First think about the direction clockwise or counterclockwise.
- It is clockwise with respect to A, so minus forces three point two into distances four meter.
- That's what minus 3 point two into 4.
- We are done with the moment of this force.
- Next, from ear to ear, we don't have any other force.
- Lastly, we have RB.
- RB, if you apply right hand thumb rule, the thumb is coming outward.
- Counterclockwise plus what is force?
- Then what is distance from B to A?
- What is the distance?
- Counted three plus four seven.
- Plus Rb into 7 is equal to 0.

- Once you simplify this, you will get the value of RB.
- To find a y we are left with one more equation sigma f y is equal to 0.
- Again start either from descent or from descent.
- Then minus 1.2 because it is downward.
- 3.2 is also downward, so minus 3.2 RB is upward, so plus RB should be equal to 0.
- Since we know the value of RB, that is 2.17, substitute that here, find the value of Ay.
- AY you will get it as 2.23 kilonewton.
- The answer is positive.
- Our assumption is correct, so a y is acting in the upward direction.
- Once you are done with this, remember reaction at A will be always equal to under root of a x squared plus a y squared.
- Since ax is 0, reaction at a will be a f itself and that will be on a y direction.
- You can go through the assignment questions, try to solve the assignment questions, and go through the question pack.
- If you come across any doubts, you can post your questions and queries in PESU forum.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 30 Beams_Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 30 Beams_Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 30 Beams_Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
CLASS 30 Beams_Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.
4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.
9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.

10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00023	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00024	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00025	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

C Revision and retrieval practice